Mapping development: A remote sensing indicator inventory for impact evaluation

Supporting evidence-informed decision-making through satellite data

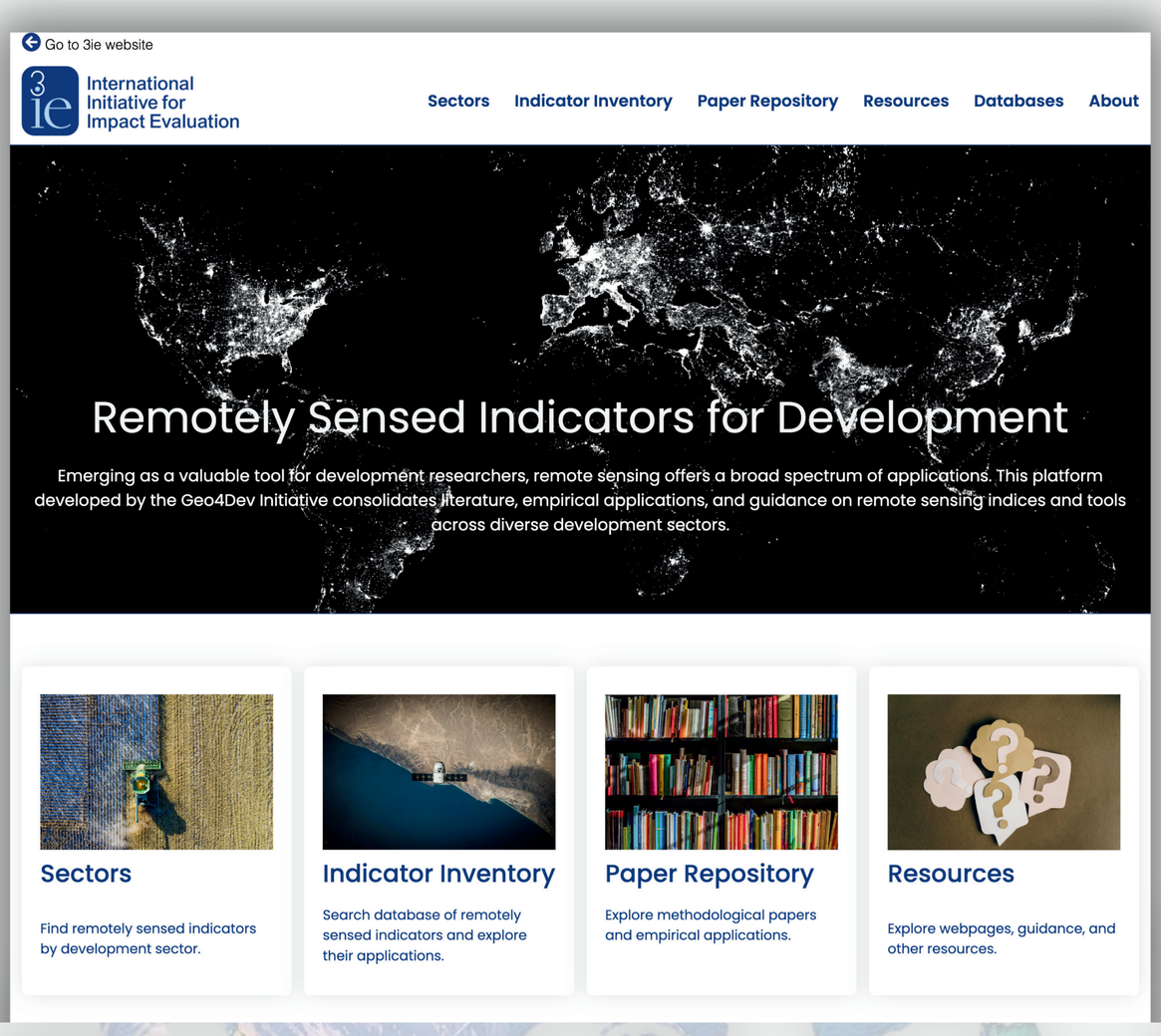
1. What is the remote sensing indicator inventory?

- A living database of indicators derived from remote sensing (RS) data
- Designed to support impact evaluation, monitoring, and decision-making
- Organized by sector, indicator type, and data availability

2. Inventory structure and contents

Three main components:

1. **Indicators by sectors:** Agriculture, Environment, Infrastructure, Health, etc.
2. **Databases:** Public datasets, satellite archives, cloud-based platforms
3. **Paper repository:** Studies using RS in evaluations, methodological papers, tagged by theme and method



3. Why this matters?

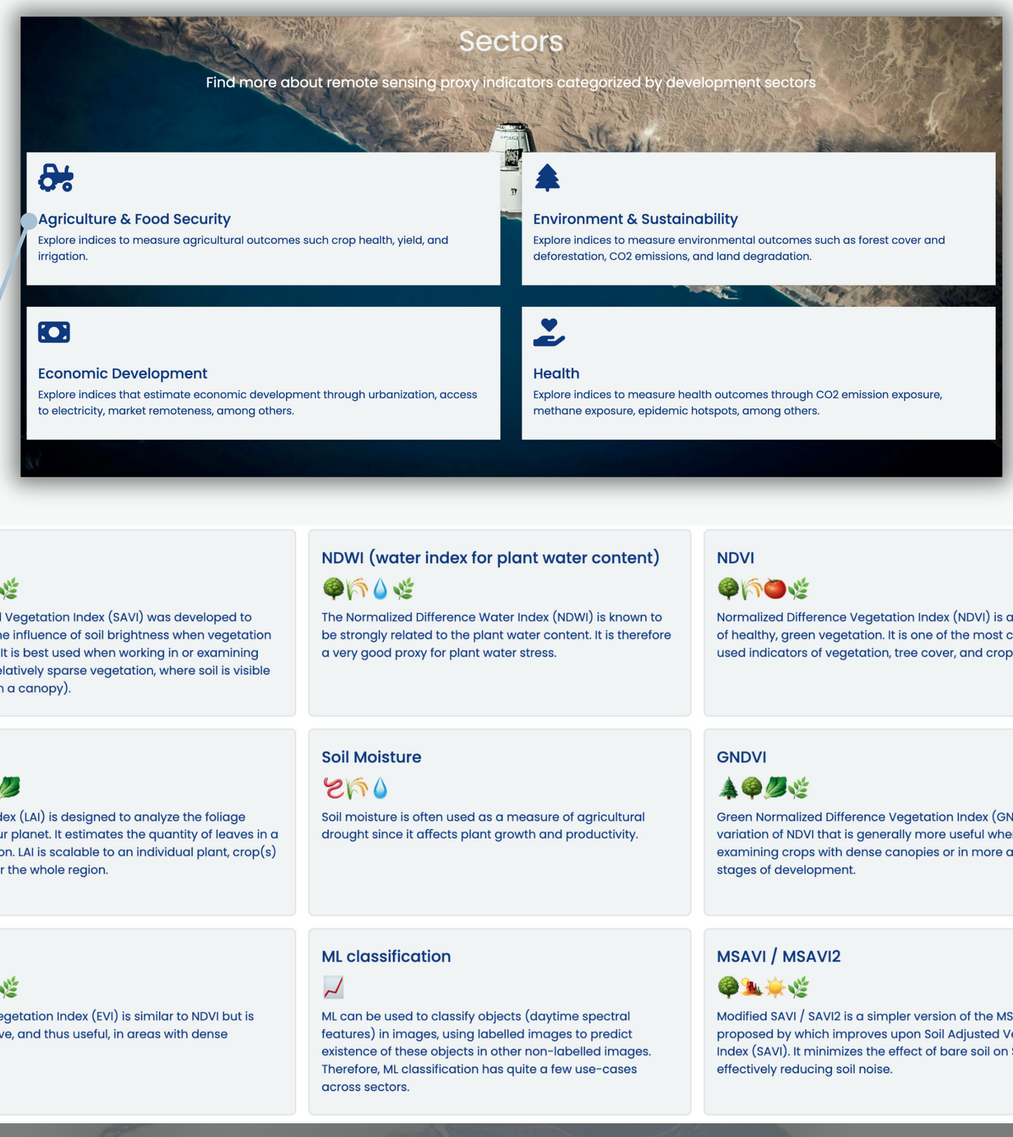
1. Makes RS data more accessible to evaluators and policymakers
2. Helps integrate geospatial evidence into impact evaluations
3. Encourages ethical, appropriate use of big data for development

4. How to access the inventory?

1. Visit: 3ieimpact.org/resources/remote-sensing-inventory
2. Explore live tool (beta): https://rsindex.softr.app

5. Why we created this inventory?

- In geospatial impact evaluations, identifying the right outcome variables and covariates is often a major challenge.
- Researchers need guidance on which RS indicators can serve as reliable proxies for development outcomes.
- To fill this gap, 3ie is developing a searchable inventory of commonly used RS indices relevant to key development sectors.
- The inventory provides:
  -Detailed descriptions of each indicator
  -Guidance on interpretation, advantages, and limitations
  -Links to studies that have used each index



6. About the team behind the inventory

- The **Remote Sensing Indicator Inventory** is developed by **3ie's Data Innovations Group (DIG)**.
- DIG guides 3ie, grantees, and partners on using big data and data science approaches ethically and effectively in impact evaluations.
- The team brings together expertise in geospatial analysis, evaluation methods, and data innovation.

International Initiative for Impact Evaluation

www.3ieimpact.org

This site represents an intermediate output from the Geo4Dev Initiative, and was developed by the International Initiative for Impact Evaluation (3ie). It aims to provide guidance on the exploration of remote sensing indices and tools across different development sectors. The Geospatial Analysis for Development, Geo4Dev, Initiative is run in collaboration between 3ie, New Light Technologies Inc., the Center for Global Action (CEGA) out of UC Berkeley.